

Zeeshan Mohammed

Data Scientist-1 (Applied CV research)

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Hyderabad, Telangana, India

SUMMARY

Applied computer vision researcher at Akridata, working on VLMs for industrial visual inspection.

EXPERIENCE

Akridata

Sep 2024 – Present

Data Scientist-I, Computer Vision R&D

Hyderabad, Telangana (Remote)

- Deployed a LoRA-tuned Florence-2 VLM in production, that boosted a medical device manufacturing client's defect-detection accuracy from 60% to 98%, while increasing inference throughput from 0.2 FPS to 4 FPS.
- Deployed a DINOv3-based few-shot ROI extractor using EoMT (encoder-only mask transformer) method
- Researched on controllable and text-guided synthetic defect generation/editing methods using diffusion models (ControlNet, AnyDoor, PowerPaint etc)
- Ran experiments for logical anomaly detection through GRPO-based VLM reasoning using frameworks like VLM-R1
- Worked on a custom, high-speed, GAN-based unsupervised anomaly detector model, delivering 60 FPS segmentation with millisecond-level latency.
- Developed an end-to-end agentic industrial visual-inspection app with a Dash UI and a FastAPI backend, integrating frontier multimodal LLMs/SLMs (API, on-device, and on-the-fly LoRA-tuning) for anomaly detection and human-in-the-loop feedback.

Akridata

May 2024 - Sep. 2024

Machine Learning Intern

Hyderabad, Telangana (Remote)

- Evaluated and tested various VLMs like Grounding DINO and OWL-ViT for zero-shot object detection
- Researched on LoRA tuning of these VLMs for few-shot object detection

IIIT-H iHub-Data | [Certificate](#)

Mar. 2024 – May.2024

Research Intern

Hyderabad, Telangana

- Interned under Prof. CK Raju. Worked on various computer vision related tasks
- Trained image segmentation models for the AutoNUE-21 dataset and tested their performance
- Used CVAT.ai to manually annotate PSU client datasets.

TECHNICAL SKILLS

Languages: Python, C++

Technologies: PyTorch, HF-Transformers, Linux, Flask, FastAPI, Dash, ChromaDB, AWS EC2/ECR/S3, Docker

PROJECTS

Fake-ViT-B/16 | [GitHub](#) | [Medium](#) | [HuggingFace](#)

PyTorch, Python, Gradio

- Replicated the Vision Transformer (2021) from Google research and fine-tuned a pre-trained checkpoint for AI-generated image detection.
- Achieved 88% test accuracy with strong generalization and deployed on HuggingFace via Gradio.

Dr.Insights | [GitHub](#) | [Medium](#)

Python, Flask, Electron.js, LangChain, Llama.cpp, ChromaDB

- Full-stack RAG+LLM medical chatbot using MSFT's Phi-2-3B, with GPT-4-Vision for analyzing lab reports and scans.
- Implemented with Flask backend, Electron.js frontend, and ChromaDB as the vector store.

Phi-2-MATH | [GitHub](#)

transformers, PEFT, QLoRA

- Fine-tuned Phi-2-3B on 40k math word problems for 500 steps on T4 GPU, achieving 0.5567 training loss.

EDUCATION

Osmania University

Hyderabad, Telangana

Bachelor of Engineering in Computer Science; CGPA: 7.75

Dec 2020 - June 2024

IIIT-Hyderabad

Hyderabad, Telangana

MS by Research in CSE (part-time)

Jan 2026 - Dec 2027